Distribution-Preserving QIM: Histogram Detectability and Robustness Trade-offs in Invisible Watermarking

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***Abstract*: Quantization index modulation (QIM) leaves a lattice fingerprint in DCT coefficients. We study Distribution-Preserving QIM (DP-QIM), which replaces a mismatched carrier with a sample from a target quantile cell rather than a lattice point. We derive an idealized cellwise mixture formula and a positive-scaling property restricted to a zero decision boundary; neither guarantees exact distribution preservation or image-level robustness. Corrected experiments use 43 crops from 14 deduplicated photographs, one outer source split for all detector fitting and calibration, and conditional source-bootstrap intervals. At 64 bits, learned-histogram detector AUC is $0.65\,[0.63, 0.76]$ for DP-QIM versus $0.95\,[0.91, 1.00]$ for QIM; DP-QIM's AUC rises to 0.84 at 256 bits and 0.96 at 1,024 bits. DP-QIM is therefore detectable, not statistically invisible. At the same DCT position and interpolated matched PSNR, it gains approximately 35, 33, and 21 percentage points in bit accuracy under blur, half-resolution downscaling, and half-contrast processing, respectively. It loses approximately 16 and 29 points under JPEG qualities 90 and 70, 14 under additive noise, and 3 to 7 under a simplified histogram-jitter attack. These small-sample results characterize a robustness/detectability trade-off, not security against strong steganalysis or the original UnMarker attack. Code and correction records are released.**[1]

***Index Terms*: invisible watermarking, quantization index modulation, DCT, steganalysis, valumetric scaling, AI-generated content provenance**

I. Introduction

Invisible watermarks are a primary tool for tracing the provenance of images, including AI-generated ones [1]. Recent work shows they are fragile against adaptive adversaries. UnMarker [2] removes deployed watermarks without access to the detector or the key by locating spectral regions whose statistics differ from those of natural images and perturbing them, and regeneration attacks remove watermarks by re-synthesizing the image [3], [4].

Quantization index modulation (QIM) [5] and its descendants are especially exposed to the first kind of attack. QIM writes each carrier coefficient onto one of two interleaved lattices, so the marginal histogram of watermarked coefficients collapses onto isolated spikes (Fig. 1, middle), a fingerprint natural images do not have. This paper asks a narrow question: *if a QIM-style watermark is constrained to leave the coefficient marginal natural, what does it gain and what does it lose?*

We study Distribution-Preserving QIM (DP-QIM). It partitions the empirical distribution of a coefficient into $2M$ equal-mass cells with alternating bit labels; a coefficient already in a cell carrying its bit is left unchanged, and otherwise it is replaced by a natural sample drawn from the interior of an adjacent cell. Our contributions are:

1) The DP-QIM embedding rule and two scoped properties: an idealized cellwise mixture formula (Prop. 1), and positive-scaling invariance at a zero boundary contrasted with periodic QIM decoding (Prop. 2).
2) A content-audited comparison protocol for quantization watermarks: both methods at the same DCT position, robustness curves compared at matched PSNR, cross-validation grouped by deduplicated source photograph, training-only detector fitting and calibration, and conditional bootstrap confidence intervals (Sec. IV).
3) An empirical characterization of the trade-off on real photographs: histogram detectability and gains under some scaling-type attacks, paid for with reduced margin against requantization and additive noise (Sec. V).

II. Related Work

QIM and its variants. QIM [5] and dither modulation are provably good under additive noise [6] but fragile to valumetric (amplitude) scaling. Rational dither modulation [7], angle QIM [8] and perceptually scaled QIM [9] restore scaling robustness by making the quantizer gain-adaptive. These schemes still write onto a (scaled) lattice; one-cell-boundary DP-QIM is scale-invariant only when that boundary is zero. Our focus is the joint empirical behaviour of histogram detectability and robustness, not a claim of new scaling invariance. We do not compare against these gain-invariant variants (Sec. VII).

Learned watermarks. HiDDeN [10], StegaStamp [11], Stable Signature [12] and InvisMark [13] train encoder-decoder networks against differentiable distortion layers. They are robust to common processing but remain vulnerable to

[1] https://github.com/sairam782/Invisible-Watermarking

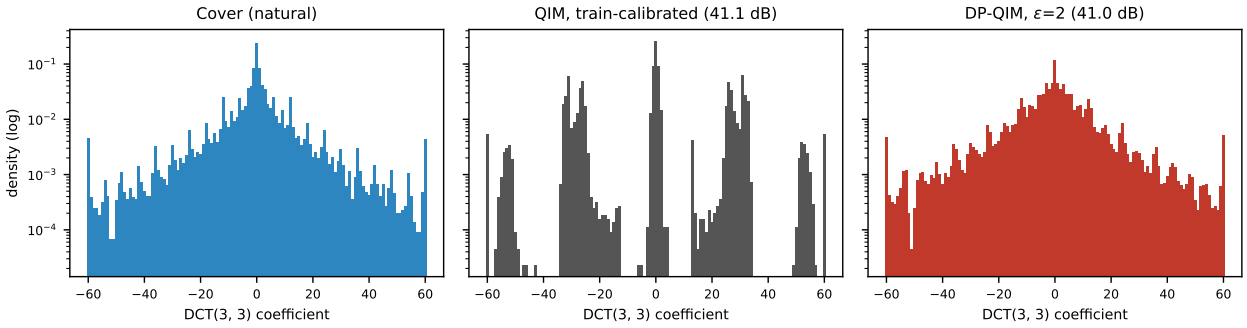


Fig. 1. Histogram (log scale) of DCT coefficient $(3, 3)$ over every block of the 43 real crops when every block carries one payload bit. Classical QIM (middle) collapses the coefficients onto lattice points; DP-QIM (right) is closer in appearance to the cover (left), without implying undetectability. QIM's step is calibrated on each outer training split; the labels report held-out PSNR, which is not forced to match exactly.

regeneration [3] and to UnMarker [2]. Analogous statistical-signature concerns arise for language-model watermarks [14].

Steganalysis and distribution preservation. Steganography seeks to hide the *existence* of a message from statistical detectors [15], [16], and distribution-preserving embedding is a classical tool there. We borrow that goal for robust watermarking, where the adversary's aim is removal rather than detection, and measure undetectability with steganalysis-style detectors.

III. Distribution-Preserving QIM

A. Setting

An RGB image is converted to YCbCr (BT.601) and the luminance is split into 8×8 blocks with orthonormal 2-D DCT. A payload of P bits is carried by one coefficient position (u, v) in P blocks chosen by a keyed permutation over interior blocks (blocks touching reflect-padding are excluded because padding is re-derived from modified pixels at decode time). Both methods below use the same position and block selection on our block-aligned crops.

DCT-QIM baseline. Dither-modulation QIM [5] with step Δ snaps the carrier c to $L_b = \{k\Delta + b\Delta/2\}$ and decodes by the nearer of L_0, L_1.

B. Embedding rule

Let F be the empirical CDF of coefficient (u, v) over a training set of natural images. For $M \geq 1$, quantile boundaries $F^{-1}(i/2M)$, $i = 0, \ldots, 2M$, define cells $\mathcal{C}_0, \ldots, \mathcal{C}_{2M-1}$ of approximately equal empirical mass with labels $\ell_i = i \bmod 2$; ties can prevent exact equality. For a safety margin $\epsilon > 0$, the *safe pool* $\mathcal{P}_i$ holds training coefficients that lie in $\mathcal{C}_i$ at least ϵ from its finite boundaries. To embed bit b in a carrier c with cell index k:

1) if $\ell_k = b$, leave c unchanged;
2) otherwise, draw one keyed sample from each adjacent pool $\mathcal{P}_{k\pm1}$ and replace c with the one closer to c.

The decoder returns the label of the cell containing the received coefficient. All experiments use $M = 1$; larger M produced narrow cells near the mode that did not survive 8-bit rounding.

C. Properties

Proposition 1 (marginal)**.** *Let $M = 1$ and let a continuous carrier distribution F be split at its median into cells C_0, C_1, each of probability $1/2$. Assume independent uniform payload bits and ideal independent sampling. Let $G_{i,\epsilon}$ be F conditioned on cell C_i and distance at least ϵ from the boundary, with positive safe mass in each cell. Then*

$$P_W = \tfrac{1}{2}F + \tfrac{1}{4}G_{0,\epsilon} + \tfrac{1}{4}G_{1,\epsilon}.$$

Writing $G_\epsilon = (G_{0,\epsilon} + G_{1,\epsilon})/2$ gives $\mathrm{KS}(P_W, F) = \frac{1}{2}\mathrm{KS}(G_\epsilon, F) \to 0$ as $\epsilon \to 0$.

Sketch. Each carrier is unchanged with probability $1/2$, independently of its value. A mismatch from each original cell occurs with probability $1/4$ and draws from the other cell's safe conditional law. This yields the mixture. Continuity implies zero mass at the median, so the two safe conditional laws converge to the original cellwise laws as the margin vanishes. Their equally weighted mixture therefore converges to F; the KS identity follows by subtracting the common component. □

Importantly, G_ϵ is not generally F conditioned globally on being outside the margin: the two cells can lose different amounts of mass. For example, safe masses in ratio $3:1$ yield global conditional weights $3/4 : 1/4$, not the sampler's $1/2 : 1/2$. This idealized result is not an exact guarantee for our capped 512-sample pools, empirical ties, or RGB rounding. For continuous F, a fixed lattice produces a different distribution, but there is no uniform positive KS lower bound as the lattice spacing tends to zero.

Proposition 2 (scaling)**.** *Let an attack map the carrier $c \mapsto s\,c$ for some $s > 0$. (i) If the $M{=}1$ cell boundary is 0, the DP-QIM decision $\mathbb{1}\{c \geq 0\}$ is unchanged for every $s > 0$. (ii) For a QIM carrier $c \in L_b$, away from decision ties, decoding preserves b iff $d_\Delta((s-1)c) < \Delta/4$, where $d_\Delta(x) = \min_{k\in\mathbb{Z}} |x - k\Delta|$. Thus $|1-s||c| < \Delta/4$ is sufficient but not necessary.*

Sketch. (i) Positive scaling preserves the two half-lines and zero. (ii) Points in the original label's lattice are separated by

Δ, and each has a decision interval of radius $\Delta/4$. Decoding depends on displacement modulo Δ, not total displacement. At equal-distance ties the code chooses bit 0. The strict bound around the original point is sufficient. □

For example, $\Delta = 16$, $c = 32$, and $s = 1/2$ preserve bit 0 even though the displacement is 16, disproving the earlier necessary-condition claim. The sign result requires an exactly zero fitted boundary; the released audit records each training-fold boundary. Blur and resizing are not generally independent scalar maps on block-DCT coefficients. Even an ideal two-dimensional Gaussian has attenuation $e^{-2\pi^2\sigma^2(f_x^2+f_y^2)}$, rather than a one-axis expression. Boundary mixing, clipping, and integer rounding require empirical evaluation; the proposition does not guarantee blur or resize robustness.

IV. Evaluation Protocol

Data. We use the real photographs distributed with scikit-image and Matplotlib (outdoor and indoor scenes, a portrait, natural textures, and astronomical and microscopy images; synthetic and text images excluded): 14 distinct sources, cut into 43 non-overlapping 256×256 crops (at most four per source). We exclude the `cat` alias of `chelsea`, deduplicate full-image hashes, and reject duplicate crop hashes. Missing sources raise errors rather than silently changing the dataset.

Splits. Five folds are grouped by *source*, so no test crop shares a photograph with any training crop. DP-QIM's CDF and pools, and the attacker's natural reference, are fitted on training sources only. Every crop is tested exactly once. Crop hashes, fold assignments, dependency versions, and calibration parameters are saved with the corrected v3 results.

Matched fidelity. Both methods embed P=64 bits at the same position (main results at $(3, 3)$). We sweep DP-QIM's margin $\epsilon \in \{0.5, 1, 2, 3, 4, 6\}$ and QIM's step $\Delta \in \{6, 8, 10, 12, 16, 20, 24, 32, 40, 48, 56, 64, 80, 96, 128\}$, and compare DP-QIM against the QIM curve linearly interpolated to DP-QIM's mean PSNR. We never extrapolate beyond the sweep. This is a descriptive comparison of held-out fidelity-robustness curves, not training-time selection of a deployed parameter. We retain $\epsilon = 2$ from the previous pilot as the main operating point; it is not selected anew to maximize the corrected results.

Attacks. JPEG (Q=90, 70), additive Gaussian noise (σ=0.03 of full range), Pillow Gaussian blur (radius 1), $0.5\times$ bicubic down-and-up-scaling, and contrast scaling about the image mean by 0.5 and 0.8 (amplitude scaling before clipping and 8-bit rounding).

Histogram-jitter removal. Inspired by statistical-anomaly attacks, we use a simplified stress test, not a reproduction of UnMarker [2]. For each of the 63 AC positions the attacker compares the received image's coefficient histogram with a natural reference built from *training-source* images, flags bins whose density exceeds the reference by a ratio τ, and adds Gaussian jitter of j times the reference inter-quartile range to coefficients in flagged bins. The attacker never sees the embedder's pools or key, but its reference and the embedder's CDF use the same training photos and are not statistically independent. We apply identical attack parameters to both methods; realized distortion can differ, and per-image attacked PSNR is recorded.

TABLE I
Crop-weighted within-fold detector ROC-AUC with conditional 95% source-bootstrap CI (0.5 = chance). Position $(3, 3)$, DP-QIM $\epsilon = 2$; QIM fidelity calibrated on training data. PSNR column reports DP-QIM.

	Learned detector		Peak statistic	
Bits (PSNR)	DCT-QIM	DP-QIM	DCT-QIM	DP-QIM
64 (53.4)	0.95 [0.91, 1.00]	**0.65 [0.63, 0.76]**	0.64 [0.52, 0.77]	**0.47 [0.41, 0.49]**
256 (46.9)	1.00 [1.00, 1.00]	**0.84 [0.79, 0.92]**	0.83 [0.67, 1.00]	**0.50 [0.39, 0.59]**
1024 (41.0)	1.00 [1.00, 1.00]	**0.96 [0.91, 1.00]**	0.97 [0.92, 1.00]	**0.62 [0.40, 0.80]**

Detectors. (a) A learned detector: logistic regression on the 1-unit-bin histogram of the carrier position over *all* blocks of an image. Following Kerckhoffs' principle it knows the method and position, but not the block-selection seed. (b) An unsupervised histogram peak statistic: the maximum over-density of the image's histogram relative to a reference built from the outer training sources. At each of five outer source folds and each $P \in \{64, 256, 1024\}$, we fit the DP-QIM CDF on training crops, construct training watermarks, and choose QIM's Δ from $\{4, 6, \ldots, 64\}$ by nearest training-mean PSNR to DP-QIM. The histogram scaler and logistic model ($C = 0.1$) are fitted on training cover/stego pairs only. We then encode and score the held-out crops without refitting or test-based calibration. Held-out PSNR can differ between methods and is reported.

Statistics. Detector ROC-AUC is calculated within each fold and averaged with held-out crop-count weights; raw scores from different fitted models are never ranked against one another. All table 95% confidence intervals use 2,000 whole-source bootstrap resamples. These intervals are conditional on the fixed fitted models and split, not uncertainty over retraining, new splits, or populations. Payload bits are seeded by CRC32 of the crop name. One fixed block-selection seed is a further limitation.

V. Results

A. Detectability

Table I shows substantial DP-QIM detectability: learned AUC increases from 0.65 to 0.84 and 0.96 as payload increases from 64 to 256 and 1,024 bits. QIM gives 0.95, 1.00 and 1.00. Thus the corrected evaluation does not support our earlier near-undetectability claim. Held-out mean PSNR is 53.36 versus 53.66 dB at 64 bits, 46.94 versus 47.12 dB at 256 bits, and 40.96 versus 41.07 dB at 1,024 bits (DP-QIM versus QIM). These residual gaps follow from training-only calibration and should not be hidden.

The fixed-direction peak score is weaker against DP-QIM (0.47, 0.50, 0.62). An AUC below 0.5 is not evidence of security: score inversion can recover discriminatory information, and the learned detector already distinguishes the classes. Finite sampling, margin exclusions, and image conversion

TABLE II
BIT ACCURACY OF DP-QIM ($\epsilon = 2$, 53.4 DB, POSITION $(3,3)$, 64 BITS) AND DIFFERENCE FROM PSNR-INTERPOLATED QIM. BOLD: CONDITIONAL 95% SOURCE-BOOTSTRAP CI EXCLUDES ZERO.

Attack	DP-QIM	Δ vs. DCT-QIM (pp)	95% CI
No attack	0.999	-0.1	[-0.2, +0.0]
JPEG Q=90	0.834	**-15.8**	[-19.8, -12.7]
JPEG Q=70	0.704	**-28.6**	[-32.7, -25.3]
Gaussian noise σ=0.03	0.830	**-14.1**	[-17.0, -11.5]
Gaussian blur r=1	0.841	**+34.8**	[+30.2, +40.2]
Downscale 0.5×	0.826	**+33.4**	[+30.0, +36.8]
Contrast ×0.5	0.967	**+20.8**	[+12.8, +32.4]
Contrast ×0.8	0.976	-0.4	[-2.6, +2.3]
Histogram jitter (τ=2, j=0.5)	0.973	**-2.7**	[-5.4, -0.9]
Histogram jitter (τ=1.5, j=1)	0.912	**-7.0**	[-10.0, -4.4]

TABLE III
GAIN (PP) OF DP-QIM ($\epsilon = 2$) OVER PSNR-INTERPOLATED QIM AT THREE DCT POSITIONS. BOLD: CONDITIONAL 95% SOURCE-BOOTSTRAP CI EXCLUDES ZERO.

Position	PSNR	Blur	Down. 0.5×	Contr. ×0.5	JPEG 70	Noise	Jitter
$(2,2)$	49.1	**+31**	**-7**	**+31**	**-16**	**-14**	**-6**
$(3,3)$	53.4	**+35**	**+33**	**+21**	**-29**	**-14**	**-7**
$(4,3)$	54.8	**+29**	**+6**	**+26**	**-37**	**-18**	**-8**

can leave detectable structure even when the histogram looks similar. We do not isolate the relative contributions of those mechanisms here.

B. Robustness at matched fidelity

Table II and Fig. 2 retain an empirical split. DP-QIM gains 34.8 pp under blur, 33.4 pp under downscaling, and 20.8 pp under half contrast; corresponding accuracies are 0.841, 0.826 and 0.967. These are measured image-level outcomes, not consequences guaranteed by Prop. 2. At contrast $\times 0.8$, the difference is -0.4 pp with an interval spanning zero. Under JPEG 90, JPEG 70 and additive noise, DP-QIM loses 15.8, 28.6 and 14.1 pp. This is consistent with less perturbation margin when sampling ordinary carrier values, but does not establish a universal ordering outside the tested data and parameters.

C. Removal stress test

The simplified histogram-jitter attack gives DP-QIM bit accuracy 0.973 ($\tau = 2, j = 0.5$) and 0.912 ($\tau = 1.5, j = 1$), respectively 2.7 and 7.0 pp below matched QIM. This does not demonstrate resistance to published UnMarker or adaptive removal. Reduced peak-score AUC does not prevent selective attacks: the learned detector distinguishes DP-QIM images.

D. Effect of position

Table III repeats the comparison at $(2,2)$ and $(4,3)$. Blur and half-contrast gains hold at the tested positions, while noise and requantization costs persist. The downscaling gain is largest at $(3,3)$, small at $(4,3)$, and negative at $(2,2)$. We do not attribute this solely to scalar attenuation because resizing mixes block content. Position dependence makes same-position comparisons necessary, but not sufficient for broad fairness.

VI. DISCUSSION

Interpretation. In this small evaluation DP-QIM lowers histogram detectability relative to QIM at low payload and improves some processing robustness measures, while losing others. It is not an undetectable watermark and should not be recommended for covert use from these results. Whether hybrids or gain-invariant QIM offer a better trade-off remains untested.

Correction record. Earlier evaluations contained frequency confounds, test-data fitting, inappropriate small-sample KS comparisons, and unstable payload seeds. The v2 audit additionally missed duplicate source aliases and inconsistent outer folds. The corrected v3 run removes the alias, validates hashes, and fits all detector components within one outer split. It corrects both the cellwise conditioning and periodic-decoding arguments. Superseded results remain in the repository for transparency; their numerical claims are withdrawn. Changes in AUC cannot be assigned to deduplication alone, because fitting and aggregation were also corrected.

VII. LIMITATIONS

- **Data and uncertainty.** Only 43 crops from 14 photographs, one fixed split and one selection seed. Conditional bootstrap intervals do not include retraining or dataset-selection variability. Hash checks remove exact duplicates, not semantic or near duplicates. Standard independent benchmarks such as Kodak or DIV2K remain necessary. These photographs were also used during method development; cross-validation is not an untouched final test.
- **Baselines.** Only lattice QIM is compared. Gain-invariant variants [7], [8], [9] and learned watermarks [10], [11] are needed. Their relative advantages cannot be inferred from this experiment.
- **Attacks and detectors.** Histogram jitter is not the published UnMarker algorithm; regeneration is not evaluated. The evaluated detectors use all-block histograms rather than exploiting the fixed public selection seed. Rich-model or CNN steganalyzers [16] may do better. One fixed noise realization per image shape is another limitation.
- **Theory and implementation.** One bit per block at one coefficient position; $M > 1$ was numerically fragile. Finite pools, empirical ties and rounding separate the implementation from the idealized theorem. Sampling conditionally on original magnitude could change both distortion and distribution and would require a new proof, not an assumption that Prop. 1 still applies.

VIII. CONCLUSION

Replacing lattice snapping with cellwise natural sampling changes the detectability/robustness trade-off, but does not make a watermark invisible. The corrected small-sample evaluation finds lower learned-histogram AUC than QIM at low payload, increasing to 0.96 for DP-QIM at full density. Gains under some processing operations coexist with losses

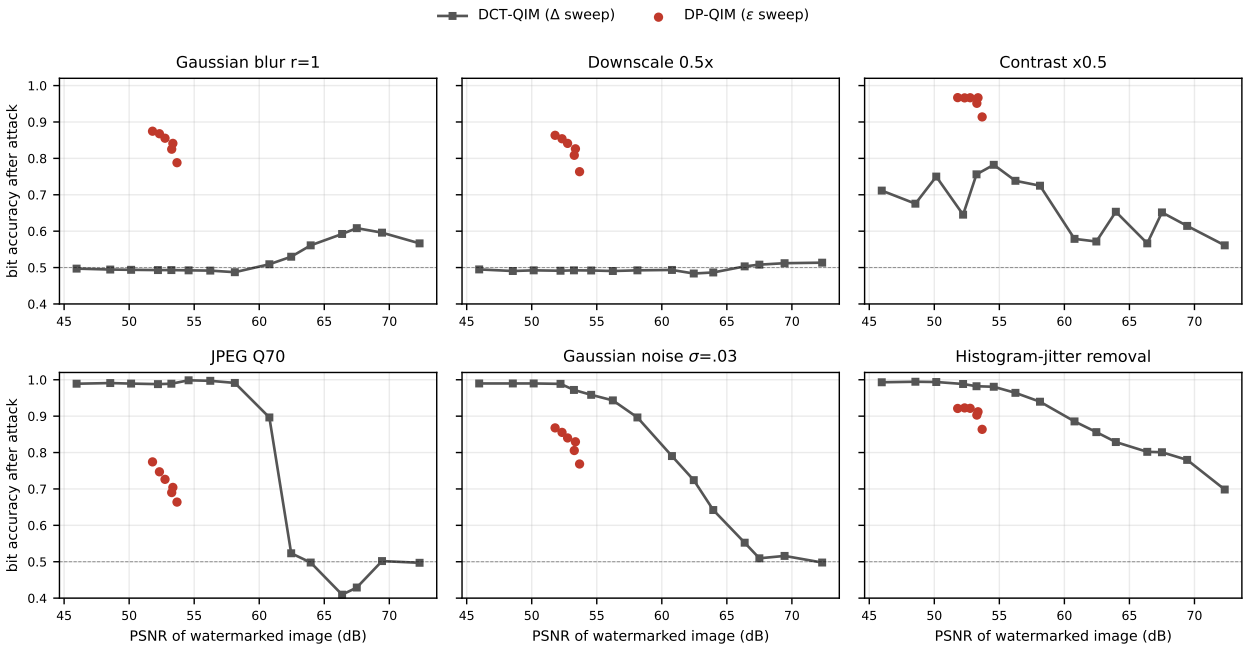


Fig. 2. Bit accuracy versus watermarked-image PSNR on 43 crops, position $(3, 3)$, 64 bits. QIM is a step-size sweep; DP-QIM points vary the safety margin. These empirical processing trade-offs are not general security guarantees.

under JPEG, noise and our simplified removal stress test. The mixture theorem is idealized, and positive-scaling invariance requires a zero boundary. Stronger baselines, independent datasets and repeated training are needed before security claims.

Reproducibility

All results are produced by the released code (https://github.com/sairam782/Invisible-Watermarking):

```
python -m experiments.eval_v2 curves --uv 3 3
python -m experiments.eval_v2 curves --uv 2 2
python -m experiments.eval_v2 curves --uv 4 3
python -m experiments.detect_v3
python -m experiments.tables_v2
python -m experiments.figures_v2
```

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